

Niharthi Muddada

HCI Researcher & Product Designer

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Work: [Research](#) | [Design Projects](#) | [Behance](#)

HCI researcher and product designer with experience studying how people navigate AI-enabled systems, healthcare experiences, and high-stakes decision-making contexts. My work combines qualitative research, systems thinking, and design to understand how patients, learners, clinicians, and service stakeholders interpret and act on complex information. I am especially interested in human-AI interaction for digital health, including how behavioral, wearable, and clinical data can be translated into trustworthy, usable, and actionable insights without replacing human judgment.

Research Interests: Human-AI interaction · Digital health · AI-supported learning · Behavior change · UX research · Multi-stakeholder systems

EDUCATION

Master of Science in Human-Centered Design & Engineering

2023 - 2025

University of Washington, United States · GPA 3.9 / 4.0 · Seattle, WA

Graduate training in human-centered research, mixed-methods inquiry, HCI, participatory design, service systems, research ethics, and translating complex human needs into interaction and technology directions.

Bachelor of Design, Industrial Design

2019 - 2023

Avantika University, India · First Class with Distinction · GPA 8.9/10 · Indore, India

Foundation in physical product design, prototyping, ergonomics, human factors, systems thinking, materials, form development, and visual communication.

SELECTED RELEVANT RESEARCH & DESIGN EXPERIENCE

2023-PRESENT

Design & Research Lead

JUL 2025 - PRESENT

Saayam for All

SEATTLE, WA

- Lead research and product design for a multi-stakeholder support platform connecting help seekers, volunteers, stewards, NGO partners, and operations teams.
- Designed the core help-request journey, task-management dashboard, and steward task-completion flow to improve how users interpret request status, priority, and ownership across the system.
- Structured information architecture and interaction workflows for a distributed service where multiple roles coordinate around the same request, reducing ambiguity in handoffs and task progression.
- Collaborate with engineering and operations on a multilingual experience across 18 languages, shaping shared interaction patterns and clearer content structures for broader accessibility.

Teaching Assistant, UX Research Methods

JAN 2025 - MAR 2025

University of Washington

SEATTLE, WA

- Supported undergraduate course instruction and student learning for a 24-student UX Research Methods class, assisting with labs, critiques, and feedback across project work.
- Helped students strengthen the connection between research questions, method choices, evidence, and design implications.
- Contributed to an academic research-learning environment by translating methodological concepts into clearer, actionable guidance.

Product Design Intern

MAY 2024 - JAN 2025

Ligi Fitness App

REMOTE

- Restructured information architecture and navigation through heuristic evaluation and a 12-participant card sort, improving usability and clarifying core workout flows.

- Designed and shipped progress-tracking and weight-moved features for an iOS fitness app, translating user needs into interaction patterns, microcopy, and high-fidelity prototypes.
- Built reusable component-library documentation and collaborated with engineering to reduce design-to-development clarification during implementation.

Social Entrepreneurship Fellow, Research & Strategy

JUN 2024 – SEP 2024

JUNO Care at the University of Washington

SEATTLE, WA

- Conducted a four-phase customer discovery study for a postpartum recovery platform, engaging mothers, clinicians, hospital administrators, and insurers to understand care needs, adoption barriers, affordability, and implementation constraints.
- Distinguished between end users and institutional buyers, identifying where product value, clinical workflow, and purchasing realities diverged.
- Synthesized findings into a more viable product and adoption direction, including provider-facing value, post-discharge engagement priorities, and lower-friction entry points through midwifery practices and community health settings.
- Communicated research outcomes through stakeholder maps, opportunity framing, strategic recommendations, and presentations for a multidisciplinary healthcare team.

Assistant Media Specialist

OCT 2023 – JUN 2024

University of Washington Continuum College

SEATTLE, WA

- Produced and updated 20+ instructional videos for university departments, ensuring educational media aligned with WCAG accessibility guidelines.
- Developed reusable visual production resources, including a font library, background-plate library, and workflow communication template, reducing friction in the team's media production process.
- Executed post-production work using Adobe Premiere Pro and After Effects, with contributions across storyboarding, motion graphics, and instructional communication.
- Supported the translation of complex educational content into clearer, more accessible visual materials for academic audiences.

UX Researcher & Product Designer

JUN 2023 – SEP 2023

Focused App

SEATTLE, WA

- Conducted user interviews with 10 participants to understand productivity needs, barriers, and expectations for a focus-oriented digital experience.
- Collaborated with a cross-functional design team to evaluate and redesign the website through heuristic review, market research, and competitive analysis.
- Generated 20+ feature concepts, including widgets and watch-app directions, translating research findings into broader product opportunities.

Product Design Intern, Research-Led Physical Product Development

JAN 2023 – JUN 2023

Dominix Global Design Studio

PUNE, INDIA

- Conducted mixed-methods research for a chronic pain management concept, combining 10+ user interviews, 40+ survey responses, and market analysis of 25+ treatment methods to identify pain-management behaviors, treatment barriers, and unmet needs.
- Translated findings into SoleSoothe, an integrated hardware-software solution combining an electronic reflexology foot therapy device with a companion mobile app for personalized treatment planning.
- Developed personas, storyboards, wireframes, product concept directions, and prototypes to communicate how physical interaction, treatment logic, and the digital experience would work together.
- Narrowed a broad solution space into a coherent product-service concept, showing how research could guide both the physical form and the supporting care experience.

SELECTED RESEARCH & DESIGN PROJECTS

Microsoft Viva / Employee Experience Platform

JAN 2025 – JUN 2025

Graduate Capstone Research · University of Washington

SEATTLE, WA

- Investigated fragmented workplace onboarding and professional learning systems through secondary research, netnography, SME interviews, and synthesis of employee experience gaps.
- Identified breakdowns across compliance onboarding, role-specific learning, and continuous development, revealing how fragmented tools and unclear support structures increase cognitive burden for new employees.
- Translated findings into a unified, context-aware support concept for Microsoft Viva, communicating the research through system maps, experience flows, and a product narrative that made complex organizational needs easier to act on.

Sutra AI/ Gyan AI

JAN 2025 – APR 2025

Product Lead / Co-Founder · AI Clinical Training Simulation

SEATTLE, WA

- Led early AI product research for an AI-supported clinical training simulation, combining interviews with medical students, competitive analysis of 20+ tools, and iterative testing to define the MVP direction.
- Shaped the product around a key learning tradeoff: supporting feedback and practice while preserving students' clinical reasoning through mandatory self-reflection before AI guidance.
- Advanced the project to the Top 16 of 174 teams in the UW Dempsey Startup Competition and secured support from the Buerk Center for Entrepreneurship, strengthening my experience in communicating research-backed product value to external reviewers and partners.

Hiya Calls Android Voicemail

JAN 2024 – APR 2024

Industry Collaboration · University of Washington / Hiya

SEATTLE, WA

- Planned and conducted usability research for an Android visual voicemail experience, developing research goals, task scenarios, participant materials, and evaluation criteria with a four-person research team.
- Synthesized findings from user sessions through affinity mapping, severity scoring, and qualitative analysis, identifying critical breakdowns in feature comprehension and interaction flow.
- Presented research insights and design recommendations to stakeholders, translating observed user friction into clear, prioritized opportunities for product improvement.

High-Altitude Water-Heating System for Mountaineers

JUN 2022 – DEC 2022

Industrial Design Project · Avantika University

INDIA

- Researched water disinfection, stove safety, and high-altitude heating challenges through market review, online failure analysis, a 21-response survey, and 5 interviews with mountaineering users.
- Defined core design requirements around safety, stability, portability, durability, and usability, then explored multiple energy and fuel concepts before selecting a solid-fuel direction through comparative evaluation.
- Designed and prototyped an integrated kettle-stove system with a plasma arc lighter, flame-regulating iris shutter, foldable ice-gripping legs, twist-lock assembly, and insulated water vessel for more reliable use in harsh mountain conditions.
- Developed concept sketches, working logic, prototype testing questions, part drawings, bill of materials, manufacturing flow, and cost estimates to carry the project from research through detailed physical design development.

COMMUNITY & SOCIAL IMPACT EXPERIENCE

Design Volunteer

APR 2022 – SEP 2022

Aashmaan Foundation

INDIA

- Supported NGO initiatives serving single and widowed parents through design and communication work, delivering 90+ campaign and outreach posters.
- Contributed to a community-focused organization addressing social and economic vulnerability, strengthening my long-term interest in design for underserved populations.

SKILLS

Research

User interviews · Mixed-methods research · Customer discovery · Secondary and literature research · Usability testing · Heuristic evaluation · Affinity mapping · Thematic analysis · Journey mapping · Research communication

Co-Design & Systems

Participatory design · Stakeholder mapping · Service and workflow design · Multi-stakeholder systems · Workshop facilitation · Inclusive design · Opportunity framing · Translating contextual insight into design direction

Physical & Communication Design

Industrial design · Physical product development · 3D concept development · Prototyping · Product-service systems · Storyboarding · Visual communication · Information architecture · Interaction design

Tools

Figma · FigJam · Miro · Adobe Creative Suite · Premiere Pro · After Effects · KeyShot · Blender · Autodesk 360

ADDITIONAL

Awards & Recognition

- Top 16 of 174 teams, University of Washington Dempsey Startup Competition — Sutra AI/ Gyan AI
- Buerk Center for Entrepreneurship support, early-stage clinical education venture — Sutra AI/ Gyan AI
- Social Entrepreneurship Fellowship, University of Washington — JUNO Care
- First Class with Distinction; Top 1% of class, B.Des. Industrial Design — Avantika University